

Co-LMLM Deletion Audit

What We Found and What We Think Is Worth Testing Next

Arya Raeesi

Hanna Roed

Working note, 25 July 2026

Short version

We ran our three-state causal audit on the public `lil-lab/CoLMLM-360M-FW` model and its Wikipedia index. From 1,500 T-REx prompts, 526 form the primary cohort: the unmodified model is correct and its selected memory entry is verified to support the answer. Under geometric-plus-value deletion, retrieval-enabled survival falls to 6.1%. Yet retrieval-disabled answerability is 26.2%, and retrieval changes a correct control answer into a wrong answer in 21.5% of cases. Deleting only the selected entry leaves 40.7% intact; geometric closure leaves 33.3%; value-aware closure leaves 7.0%. Value deletion is effective but broad, removing a median of 21 and mean of 57.3 entries per target. A verbatim answer-bearing entry placed just outside the geometric radius also restores many answers. Our read is that the main open problem is the unit of deletion: a cosine neighborhood, answer string, source, and database row are not the same as a fact identity. Identity clustering is not implemented here. What we have now is the audit layer and a set of baselines for testing it.

1 What we ran, and why

Co-LMLMexternalizes factual spans into a continuous-key, text-value index and retrieves the top item using a hidden-state query [1].

The paper already shows effective aggregate forgetting on TOFU [2]. We wanted to zoom in on the mechanism behind that result. The narrower question is: once a target entry is removed from a schema-free index, does a correct answer come from another memory entry, or from the model and prompt without factual retrieval? That distinction tells us whether the next useful intervention belongs in the deletion policy, the retrieval policy, or the model.

Our design adapts the causal framework for relational LMLMs [3]. For each fact, the model, prompt, decoding, and parameters are fixed while memory access changes:

State	Intervention
FULL	Memory unchanged; retrieval enabled.
DEL-ON	Target deletion set hidden; retrieval enabled.
DEL-OFF	Target deletion set hidden; factual retrieval disabled.

Deletion uses reversible search-time filtering rather than physically rewriting the index. Let $C_s(f)$ indicate whether the output in state s contains the normalized gold answer or an accepted alias. Conditional on FULL correctness, we report:

$$L(f) = C_{\text{DEL-OFF}}(f), \quad \text{retrieval-disabled answerability,} \quad (1)$$

$$R(f) = C_{\text{DEL-ON}}(f)(1 - C_{\text{DEL-OFF}}(f)), \quad \text{retrieval-mediated correctness,} \quad (2)$$

$$I(f) = (1 - C_{\text{DEL-ON}}(f))C_{\text{DEL-OFF}}(f), \quad \text{retrieval interference,} \quad (3)$$

$$S(f) = C_{\text{DEL-ON}}(f), \quad \text{post-deletion survival.} \quad (4)$$

We additionally inspect DEL-ON traces for retained candidates that support the target. Trace support is evidence available to the model, but its absence does not prove weight memorization; a correct answer can also come from the prompt or general language regularities.

Table 1: Construction of the auditable cohort.

Stage	Count	Coverage
All T-REx prompts	1,500	100.0%
FULL produced a selected entry ID	1,190	79.3%
Selected entry passed the target-support judge	599	39.9%
Support verified and FULL answer correct	526	35.1%

The unmodified model is correct on 763 of 1,500 prompts (50.9%). The primary causal cohort is the 526 cases that are both FULL-correct and support-verified. The remaining 901 support exclusions comprise 310 prompts with no selected entry ID and 591 whose selected entry fails the support judge. All main survival rates are conditional on the 526-case cohort.

We use greedy decoding with at most 12 new tokens on `lil-lab/CoLMLM-360M-FW`. The standard policy combines geometric closure at $\rho = 0.85$ with value support judged inside a top-500 retrieval envelope and at run time. The completed closures are not truncated. The run uses the public, approximate Wikipedia index (approximately 240M entries), not the 2.2B-entry Wikipedia-FineWeb-Edu index [1]. The estimates therefore describe the released end-to-end system, including approximate search.

2 What we found

2.1 Channel decomposition

Table 2: Paired outcomes under geometric-plus-value deletion at $\rho = 0.85$, restricted to the 526-case primary cohort.

Outcome	Count	Rate
Wrong in both DEL-ON and DEL-OFF	381	72.4%
Correct only in DEL-ON: retrieval-mediated R	7	1.3%
Correct only in DEL-OFF: retrieval interference I	113	21.5%
Correct in both post-deletion states	25	4.8%
Any DEL-ON correctness: survival S	32	6.1%
Any DEL-OFF correctness: L	138	26.2%

The standard policy leaves 32 of 526 answers intact (6.1%, Wilson 95% interval [4.3, 8.5]), but only seven (1.3%) are correct uniquely with retrieval. In contrast, 138 (26.2%) are correct without factual

retrieval, and in 113 (21.5%) a surviving retrieval changes that correct control answer into a wrong answer. All 32 DEL-ON-correct traces lack retained target-supporting evidence; 25 are nevertheless also correct in DEL-OFF. Thus 6.1% should not be called retrieval leakage: the retrieval-mediated subset is 1.3%.

The DEL-OFF result is stable under a stricter control that forbids the retrieval token: 132 of 526 are correct (25.1%), versus 138 (26.2%) under null retrieval. The paired difference is 1.1 percentage points ($p = 0.377$). We call this *retrieval-disabled answerability*, not pure weight memorization, because prompts and general language knowledge remain available.

2.2 Deletion policy and scope

We compare several possible meanings of “delete this fact.” Selected-entry deletion hides only the verified row used in FULL; provenance deletion hides its source; geometric deletion hides keys with cosine similarity at least 0.85; value deletion hides retrieved entries judged to support the target answer; and the combined policies take their union. These are baselines, not identity clusters.

Table 3: Policy comparison on the 526-fact cohort. Closure sizes are measured over all 599 support-verified targets. Provenance deletion is source-wide, so its true hidden-entry count is not available from the saved candidate list. Efficacy is one minus survival; “gold in trace” is the number of survivors with retained target-supporting evidence.

Policy	Survival	Efficacy	Gold in trace	Mean	Median	Max.
Selected entry	40.7%	59.3%	192/214	1	1	1
Provenance	41.6%	58.4%	219/219	—	—	—
Geometric, $\rho = 0.85$	33.3%	66.7%	154/175	2.5	1	183
Value	7.0%	93.0%	0/37	57.3	21	494
Geometric + value	6.1%	93.9%	0/32	57.9	21	494
Hybrid	5.7%	94.3%	0/30	58.4	22	494

Removing the selected supporting entry leaves 214 answers intact, and 192 of those survivors retrieve other target-supporting evidence. Our read is that deleting one observed row is not yet the same as deleting the fact. The same record–fact mismatch has been identified as a structural challenge for selective forgetting in RAG systems [4]. Geometry removes slightly more, but 154 geometric survivors still have gold evidence in their traces. Value matching drives the large improvement: 141 cases survive geometry but not value, versus three in the reverse direction ($p < 5 \times 10^{-38}$).

The tradeoff is broad deletion. Value closure removes a median of 21 and mean of 57.3 observed entries per target, reaching 494 within a 500-candidate envelope. Common answers can belong to unrelated propositions, so value closure is a strong efficacy baseline rather than a precise identity policy. Adding geometry or provenance after value changes survival only marginally.

2.3 Geometric boundary stress test

For each target, this stress test inserts one answer-bearing entry at similarity $\rho - \epsilon$, just outside the geometric deletion radius $\rho = 0.85$. The baseline is geometric deletion alone, whose survival rate is 33.3%.

Table 4: Verbatim survivor placed outside the geometric radius. Attack gain is the fraction changed from wrong under baseline deletion to correct.

ϵ	Entry selected	Post-attack correct	Attack gain
0.01	87.5%	68.6%	40.9%
0.02	77.8%	67.1%	37.8%
0.05	57.6%	57.8%	27.4%

An exact answer-bearing value immediately outside the radius is often selected and restores the answer. We therefore see the cosine threshold as a routing boundary, not yet a semantic deletion guarantee. Hyphenated, letter-spaced, and prefix-cued values produce almost no new correct answers (attack gain $\leq 0.4\%$). This run supports a verbatim boundary-evasion claim, not a general paraphrase or obfuscation claim.

2.4 Secondary diagnostics

Representation probe. A five-fold, fact-disjoint linear probe predicts an accepted answer from frozen FULL query vectors for 139 of 599 facts (23.2%), compared with 6.8% for the most common answer. The overlap with behavioral DEL-OFF correctness is weak (Jaccard $\approx 18.3\%$). Query vectors therefore contain answer-predictive structure, but this is not evidence that they store complete facts or the same facts recovered behaviorally.

Radius sweep and collateral. Geometric efficacy rises from 60.1% at $\rho = 0.95$ to 78.9% at $\rho = 0.70$; measured collateral reaches only 0.20%. We do not treat the collateral number as conclusive. Of 526 target neighborhoods, 476 had to be filled with facts outside the requested cosine ball, and only 564 of 2,747 evaluated links were actually inside it. The entanglement gap is also degenerate on this grid (469 zeros and 57 ones), while behavior is non-monotone as deletion exposes different candidates. The neighborhood definition must be redesigned before using this as a headline result.

Margin predictor. We could not estimate margin AUROC in this run. The geometric search stops at ρ , so the best sub-threshold survivor score is missing. A threshold-free retrieval envelope is required before this diagnostic is valid.

3 What we think is worth testing next

Let each memory entry be $e_i = (k_i, v_i, s_i)$, containing a continuous key, text value, and source metadata. We think the next strong baseline is a persistent identity assignment z_i such that

$$z_i = z_j \iff e_i \text{ and } e_j \text{ express the same deletable proposition,}$$

under an explicit policy. A deletion request for identity z should hide every e_i with $z_i = z$, while retaining entries that merely share an answer, topic, or source.

For example, “X was born in Paris” and “Y was born in Paris” share an answer but are different propositions, while “X’s birthplace is Paris” and “X was born in the French capital” may be equivalent. Identity must therefore combine retrieval geometry with text and provenance rather than use unconstrained clustering.

3.1 A minimal identity-aware baseline

1. Generate high-recall candidate pairs using key neighbors, value overlap, entities, and source links.
2. Label a benchmark with *same proposition*, *same answer only*, *related*, and *unrelated* pairs. Shared-answer cases are essential hard negatives.
3. Score proposition equivalence using bidirectional entailment, entity/relation consistency, context, provenance, and key similarity. Form constrained clusters and store an inspectable, persistent identity ID.
4. Delete atomically by identity and run the existing three-state audit without changing the evaluation cohort or decoding.

The comparison we would care about is concrete: can this approach get close to value closure’s 93.0% efficacy while deleting far fewer than its mean of 57.3 entries, with high human-validated cluster precision and low behavioral collateral? We would keep selected-entry, provenance, geometric, value, geometric-plus-value, hybrid, and a small manually verified semantic oracle as the baselines. We would report L , R , I , trace evidence, cluster precision/recall, cluster size, latency, and collateral on separate hard-negative sets for shared answer, entity, relation, source, and key neighborhood. Splits should be identity-disjoint.

Separately, two follow-ups look useful. First, calibrated abstention could help when deletion leaves only weak or conflicting evidence; the interference rate shows that blindly accepting the next candidate can hurt. Second, we should repair the margin and collateral diagnostics, extend the boundary test to natural paraphrases and distributed copies, and then repeat the locked protocol on the 2.2B-entry combined index.

4 What this note does not claim

To keep the interpretation clean:

- Results are conditional on 526 FULL-correct, support-verified T-REx prompts, not all knowledge in the model or index.
- Correctness is normalized answer containment; the automated support judge and value predicate still require human validation.
- DEL-OFF removes factual retrieval but does not separate parameters from information in the prompt.
- Search-time filtering tests behavioral deletion, not physical erasure, replicas, caches, or later index rebuilds.
- Approximate search, the 500-candidate value envelope, and the observed query limit any claim of exhaustive semantic deletion.
- Identity clustering, modified pretraining, and learned abstention have not yet been implemented.

5 Our overall read

Our overall read is positive for the Co-LMLM direction: the external memory is editable, inspectable, and clearly doing real work. The part that still looks open is the semantic unit of deletion. Selected-entry and geometric policies miss many alternative answer-bearing spans; value-aware deletion closes

most paths but is too broad to serve as an identity rule. We now have the paired interventions, traces, controls, and baselines needed to test the next step together: proposition-level identity clusters that support atomic deletion with measured efficacy and collateral damage.

Statistical note. Intervals are two-sided 95% Wilson intervals. Paired comparisons use two-sided exact McNemar tests; exploratory tests are not adjusted for multiplicity.

References

- [1] Yair Feldman, Linxi Zhao, Nathan Godey, Dongyoung Go, Yilun Hua, Kilian Q. Weinberger, Jennifer J. Sun, and Yoav Artzi. Co-LMLM: Continuous-Query Limited Memory Language Models. *arXiv preprint arXiv:2607.07707*, 2026. <https://arxiv.org/abs/2607.07707>.
- [2] Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. TOFU: A Task of Fictitious Unlearning for LLMs. *arXiv preprint arXiv:2401.06121*, 2024. <https://arxiv.org/abs/2401.06121>.
- [3] Arya Raeesi and Hanna Roed. Auditing Forgetting in Limited Memory Language Models. *arXiv preprint arXiv:2607.00605*, 2026. <https://arxiv.org/abs/2607.00605>.
- [4] Leila Tavakoli and Mark Sanderson. Deletion Isn’t Enough: Auditing RAG for Selective Forgetting. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR ’26)*, pages 2721–2733, 2026. <https://doi.org/10.1145/3805712.3808545>.